

**A WAITING TIME DISTRIBUTION ARISING FROM THE
COUPON COLLECTOR'S PROBLEM**

BY

HERMAN RUBIN and JAMES ZIDEK

TECHNICAL REPORT NO. 107

AUGUST 5, 1965

**THIS RESEARCH WAS SPONSORED BY THE ARMY RESEARCH OFFICE,
OFFICE OF NAVAL RESEARCH, AND AIR FORCE OFFICE OF
SCIENTIFIC RESEARCH BY CONTRACT NO.
Norr-225(52) (NR 342-022)**

**DEPARTMENT OF STATISTICS
STANFORD UNIVERSITY
STANFORD, CALIFORNIA**



A WAITING TIME DISTRIBUTION ARISING FROM THE
COUPON COLLECTOR'S PROBLEM

by

Herman Rubin and James Zidek

TECHNICAL REPORT NO. 107

August 5, 1965

PREPARED UNDER CONTRACT Nonr-225(52)

(NR 342-022)

FOR

OFFICE OF NAVAL RESEARCH

Reproduction in Whole or in Part is Permitted for
any Purpose of the United States Government

DEPARTMENT OF STATISTICS
STANFORD UNIVERSITY
STANFORD, CALIFORNIA

A WAITING TIME DISTRIBUTION ARISING FROM THE
COUPON COLLECTOR'S PROBLEM

by

Herman Rubin and James Zidek

Summary

This report is concerned with a certain waiting time distribution which arises in the context of the Coupon Collector's Problem. In general, the term "Coupon Collector's Problem" refers to the problem which confronts an individual who is attempting to collect a subset of some set of distinct items, by sampling with replacement, one item from this set on each of a sequence of independent trials, until this subset is obtained. The waiting time in question, denoted by $S(r, N)$ ($N = 1, 2, \dots$; $r = 1, 2, \dots, N$), is the number of trials needed to obtain for the first time, r distinct elements from among N distinct elements, if each of the sequence of independent trials consists of sampling with replacement and equal probabilities, one element from among the N .

In Section 1 of this report, the exact distribution of $S(r, N)$ is obtained. Using this result, we derive in the second section, an asymptotic approximation to this distribution for the case $r = N$. Calculations illustrating the effectiveness of this approximation in the case $N = 1000$, are given.

1. The Exact Distribution of $S(r, N)$

Observe that $S(r, N)$ may alternatively be defined in the following way. On each of a succession of independent trials, a ball is placed at random in any one of N urns, each urn being equally likely to receive the ball. This procedure is continued until exactly $N - r$ urns remain empty. The number of trials so required is $S(r, N)$.

With this interpretation, the distribution of $S(r, N)$ is readily obtained using a probabilistic argument. Suppose the urns are numbered from 1 to N . Let A_{jn} denote the event that a ball is placed in urn j at trial n for the first time and $E_j(m, n)$, the event that after trial n , exactly m among the urns excluding the j^{th} remain empty ($j=1, 2, \dots, N$; $m=0, 1, 2, \dots, N-1$; $n=1, 2, \dots$). Define $g(r, N, s)$ by

$$(1) \quad g(r, N, s) = P[S(r, N) = s] \quad , \quad (s = r, r + 1, \dots) \\ = 0 \quad , \quad s < r \quad .$$

Then

$$(2) \quad g(r, N, s) = \sum_{j=1}^N P[A_{js} E_j(N - r, s - 1)] \\ = N P[A_{1s} E_1(N - r, s - 1)] \\ = N \cdot \frac{1}{N} \cdot \left(\frac{N-1}{N} \right)^{s-1} P[E_1(N - r, s - 1) | A_{1s}] \quad .$$

The conditional probability in equations (2), being just the probability that exactly $N - r$ among $N - 1$ urns remain empty after $s - 1$ balls are distributed at random among the urns, is known and in Feller's^{1/} notation is just $p_{N-r}(s-1, N-1)$.

^{1/} Feller, W., An Introduction to Probability Theory and its Applications, Volume 1, Second Edition, New York: Wiley, 1962, p. 92.

Thus

$$g(r, N, s) = \binom{N-1}{N-r} \left(\frac{1}{N}\right)^{s-1} \sum_{k=0}^{r-1} (-1)^k \binom{r-1}{k} (r-1-k)^{s-1}$$

(3) $s = r, r+1, \dots$

$$= 0 \quad s < r$$

If $G(r, N, s)$ denotes the distribution function corresponding to $g(r, N, s)$, then

$$1 - G(r, N, s) = P[S(r, N) > s]$$

(4) $= \sum_{k=s+1}^{\infty} g(r, N, k)$

Thus,

$$1 - G(r, N, s) = \binom{N-1}{N-r} \sum_{k=0}^{r-1} (-1)^k \binom{r-1}{k} \left(\frac{r-1-k}{N}\right)^s$$

(5) $\times \left[\frac{N}{N-r+1+k} \right], \quad s = r, r+1, \dots$

$$= 1, \quad s < r$$

When $r = N$, equation (5) yields after simplification

$$1 - G(N, N, n) = \sum_{k=1}^N (-1)^{k-1} \binom{N}{k} \left(1 - \frac{k}{N}\right)^n \quad n = N, N+1, \dots$$

(6) $= 1, \quad n < N$

This last result is given in Feller (see the previous reference, p. 59).

2. An Asymptotic Approximation to $G(N, N, n)$

Suppose, in $G(N, N, n)$, we set $\frac{1}{2}$

$$(7) \quad n = N(\log N + q) + \frac{1}{2},$$

where q is regarded as fixed. Recall that in general

$$(8) \quad m! \sim \sqrt{2\pi} m^{m+1/2} \exp\left\{-m + \frac{1}{12m} - \frac{1}{360m^3}\right\}$$

as $m \rightarrow \infty$ and

$$(9) \quad -\log(1-u) = u + uv + \frac{2}{3} v^3 + \frac{2}{5} v^5 + \dots, \quad |u| < 1,$$

where $v = u/(2-u)$. The expansion of equation (9) is rapidly convergent for small values of $|u|$ and hence more useful, for our purposes than the usual expansion

Using equation (8) we have

$$\begin{aligned} \log \left[\frac{N!}{(N-k)!} \right] &\sim (N + \frac{1}{2}) \log N - (N - k + \frac{1}{2}) \log(N - k) \\ &- \frac{1}{12} \left(\frac{1}{N} - \frac{1}{N-k} \right) - \frac{1}{360} \left(\frac{1}{N^3} - \frac{1}{(N-k)^3} \right) \end{aligned}$$

and hence

$\frac{1}{2}$

For convenience we shall write $\log x$ for $\log_e x$.

$$(10) \quad \log \left[\frac{N!}{(N-k)!} \right] \sim k \log N - (N-k + \frac{1}{2}) \log(1 - \frac{k}{N}) - k \\ + \frac{1}{12} \left(\frac{1}{N} - \frac{1}{N-k} \right) - \frac{1}{360} \left(\frac{1}{N^3} - \frac{1}{(N-k)^3} \right) ,$$

as $N \rightarrow \infty$. At the same time, from equation (9)

$$(11) \quad \log(1 - \frac{k}{N})^n = [N(\log N + q) + \frac{1}{2}] \left[-\frac{k}{N} - \frac{k}{N} v - \frac{2}{3} v^3 \dots \right] ,$$

where

$$(12) \quad v = \frac{k}{N} / (2 - \frac{k}{N}) .$$

Combining equations (10) and (11), we obtain upon simplification

$$(13) \quad \log \left[\frac{N!}{(N-k)!} (1 - \frac{k}{N})^n \right] \sim -kq + k \log(1 - \frac{k}{N}) \\ + N[\log N + q - 1] \left[\log(1 - \frac{k}{N}) + \frac{k}{N} \right] \\ + \frac{1}{12} \left(\frac{1}{N} - \frac{1}{N-k} \right) - \frac{1}{360} \left(\frac{1}{N^3} - \frac{1}{(N-k)^3} \right) .$$

From expansion (13) and equation (6), it follows that as $N \rightarrow \infty$

$$(14) \quad 1 - G(N, N, n) \sim \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{r!} e^{-kq} X_N ,$$

where

$$\begin{aligned}
(15) \quad X_N &= \exp\left\{k \log\left(1 - \frac{k}{N}\right) + N[\log N + q - 1]\left[\log\left(1 - \frac{k}{N}\right) + \frac{k}{N}\right] \right. \\
&\quad \left. + \frac{1}{12} \left(\frac{1}{N} - \frac{11}{N-k}\right) - \frac{1}{360} \left(\frac{1}{N^3} - \frac{1}{(N-k)^3}\right) \right\} .
\end{aligned}$$

Observe that if in equation (15) we expand $\log(1 - \frac{k}{N})$ using equation (9) and in turn represent the exponential by its series, that by interchanging the orders of summation in the sum of (14), we obtain an asymptotic expansion of $1 - G(N, N, n)$ in powers of $1/N$, that is,

$$\begin{aligned}
(16) \quad 1 - G(N, N, n) &\sim 1 - \exp(-e^{-q}) \\
&\quad + \left[\frac{1 + \log N + q}{2N} \right] e^{-q}(e^{-q} - 1) \exp(-e^{-q}) + \dots ,
\end{aligned}$$

since

$$(17) \quad \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k!} e^{-kq} = 1 - \exp(-e^{-q})$$

and

$$\begin{aligned}
(18) \quad \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k!} e^{-kq} k^2 &= \left(x \frac{d}{dx} \right)^2 e^{-x} \Big|_{x=e^{-q}} \\
&= \exp(-e^{-q}) e^{-q} (1 - e^{-q}) .
\end{aligned}$$

Thus

$$(19) \quad 1 - G(N, N, n) \sim 1 - \exp(-e^{-q}) .$$

For a given value of q the error, say $E_N(q)$, in $1-G(N,N,n)$ resulting from the use of the approximation given by (19) is, asymptotically as $N \rightarrow \infty$,

$$(20) \quad E_N(q) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k!} e^{-kq} [X_N - 1] .$$

For all real x , let $[x]$ denote the closest integer to x . Denote by n_p the integer closest to the p^{th} percentile of the distribution whose distribution function is given by

$$(21) \quad \exp \left\{ -e^{-\frac{y-1/2}{N} + \log N} \right\} \quad (-\infty < y < \infty) .$$

Then

$$(22) \quad n_p = [N(\log N - \log \log p^{-1}) + \frac{1}{2}]$$

and n_p lies in the proximity of the p^{th} percentile for G .

As an indication of the accuracy of the approximation given by (19), $E_N(q_p)$, where $q_p = -(n_p - 1/2) / N + \log N$, has been computed for the cases $p = 0.001, 0.01, 0.1, 0.9, 0.99$ and 0.999 and $N = 1000$. We have also obtained, in these cases, the values of the quantity $E_N'(q_p)$ defined by

$$(23) \quad E'_N(q_p) = \left(\frac{1 + \log N + q_p}{2N} \right) e^{-q_p} (e^{-q_p} - 1) \exp(-e^{-q_p})$$

which is {see (16)} asymptotically equal to $E_N(q_p)$. These results together with the corresponding values of n_p are given in the accompanying table.

VALUES OF n_p FOR VARIOUS VALUES OF p , $N = 1000$, AND THE CORRESPONDING VALUES OF $E_N(q_p)$, $E'_N(q_p)$

p	n_p	$E_N(q_p)$	$E'_N(q_p)$
.001	4976	$1.18973 \ 3533 \times 10^{-4}$	1.2215×10^{-4}
.01	5381	$5.28372 \ 5699 \times 10^{-4}$	5.2957×10^{-4}
.1	6074	$1.06769 \ 7178 \times 10^{-3}$	1.0609×10^{-3}
.9	9159	$-4.30581 \ 8673 \times 10^{-4}$	-4.3077×10^{-4}
.99	11508	$-6.14536 \ 4900 \times 10^{-5}$	-6.1593×10^{-5}
.999	13816	$-7.37067 \ 4802 \times 10^{-6}$	-7.3930×10^{-6}

Unclassified

Security Classification

DOCUMENT CONTROL DATA - R&D		
(Security classification of title, body of abstract and indexing annotation must be entered when the overall report is classified)		
1. ORIGINATING ACTIVITY (Corporate author) Department of Statistics Stanford University Stanford, California		2a. REPORT SECURITY CLASSIFICATION Unclassified
		2b. GROUP
3. REPORT TITLE A Waiting Time Distribution Arising From the Coupon Collector's Problem		
4. DESCRIPTIVE NOTES (Type of report and inclusive dates) Technical Report		
5. AUTHOR(S) (Last name, first name, initial) Rubin, Herman and Zidek, James, V.		
6. REPORT DATE August 5, 1965	7a. TOTAL NO. OF PAGES 8	7b. NO. OF REFS 1
8a. CONTRACT OR GRANT NO. Nonr 225(52)	9a. ORIGINATOR'S REPORT NUMBER(S) Technical Report No. 107	
b. PROJECT NO. NR 342-022		
c.	9b. OTHER REPORT NO(S) (Any other numbers that may be assigned this report)	
d.		
10. AVAILABILITY/LIMITATION NOTICES Releaseable without limitations on dissemination		
11. SUPPLEMENTARY NOTES	12. SPONSORING MILITARY ACTIVITY Logistics and Mathematical Sciences Branch Office of Naval Research Washington, D.C. 20360	
13. ABSTRACT This report is concerned with a certain waiting time distribution which arises from the Coupon Collector's Problem. In general, "Coupon Collector's Problem" refers to the problem which confronts an individual who is attempting to collect a subset of some set of distinct items by sampling with replacement and equal probabilities, one item from this set on each of a sequence of independent trials until this subset is obtained. The number of trials so required is the waiting time, S, in question. In the first portion of this report, the exact distribution of S is determined. In the second, an approximation to this distribution is derived in the case where the subset is the set itself. Calculations illustrating the effectiveness of this approximation are given.		

Unclassified

Security Classification

Unclassified
Security Classification

14. KEY WORDS	LINK A		LINK B		LINK C	
	ROLE	WT	ROLE	WT	ROLE	WT
Coupon Collecting Coupon Collector's Problem Waiting Times in Coupon Collecting Distribution Theory for an Urn Model						

INSTRUCTIONS

1. ORIGINATING ACTIVITY: Enter the name and address of the contractor, subcontractor, grantee, Department of Defense activity or other organization (*corporate author*) issuing the report.

2a. REPORT SECURITY CLASSIFICATION: Enter the overall security classification of the report. Indicate whether "Restricted Data" is included. Marking is to be in accordance with appropriate security regulations.

2b. GROUP: Automatic downgrading is specified in DoD Directive 5200.10 and Armed Forces Industrial Manual. Enter the group number. Also, when applicable, show that optional markings have been used for Group 3 and Group 4 as authorized.

3. REPORT TITLE: Enter the complete report title in all capital letters. Titles in all cases should be unclassified. If a meaningful title cannot be selected without classification, show title classification in all capitals in parenthesis immediately following the title.

4. DESCRIPTIVE NOTES: If appropriate, enter the type of report, e.g., interim, progress, summary, annual, or final. Give the inclusive dates when a specific reporting period is covered.

5. AUTHOR(S): Enter the name(s) of author(s) as shown on or in the report. Enter last name, first name, middle initial. If military, show rank and branch of service. The name of the principal author is an absolute minimum requirement.

6. REPORT DATE: Enter the date of the report as day, month, year, or month, year. If more than one date appears on the report, use date of publication.

7a. TOTAL NUMBER OF PAGES: The total page count should follow normal pagination procedures, i.e., enter the number of pages containing information.

7b. NUMBER OF REFERENCES: Enter the total number of references cited in the report.

8a. CONTRACT OR GRANT NUMBER: If appropriate, enter the applicable number of the contract or grant under which the report was written.

8b, 8c, & 8d. PROJECT NUMBER: Enter the appropriate military department identification, such as project number, subproject number, system numbers, task number, etc.

9a. ORIGINATOR'S REPORT NUMBER(S): Enter the official report number by which the document will be identified and controlled by the originating activity. This number must be unique to this report.

9b. OTHER REPORT NUMBER(S): If the report has been assigned any other report numbers (*either by the originator or by the sponsor*), also enter this number(s).

10. AVAILABILITY/LIMITATION NOTICES: Enter any limitations on further dissemination of the report, other than those

imposed by security classification, using standard statements such as:

- (1) "Qualified requesters may obtain copies of this report from DDC."
- (2) "Foreign announcement and dissemination of this report by DDC is not authorized."
- (3) "U. S. Government agencies may obtain copies of this report directly from DDC. Other qualified DDC users shall request through _____."
- (4) "U. S. military agencies may obtain copies of this report directly from DDC. Other qualified users shall request through _____."
- (5) "All distribution of this report is controlled. Qualified DDC users shall request through _____."

If the report has been furnished to the Office of Technical Services, Department of Commerce, for sale to the public, indicate this fact and enter the price, if known.

11. SUPPLEMENTARY NOTES: Use for additional explanatory notes.

12. SPONSORING MILITARY ACTIVITY: Enter the name of the departmental project office or laboratory sponsoring (*paying for*) the research and development. Include address.

13. ABSTRACT: Enter an abstract giving a brief and factual summary of the document indicative of the report, even though it may also appear elsewhere in the body of the technical report. If additional space is required, a continuation sheet shall be attached.

It is highly desirable that the abstract of classified reports be unclassified. Each paragraph of the abstract shall end with an indication of the military security classification of the information in the paragraph, represented as (TS), (S), (C), or (U).

There is no limitation on the length of the abstract. However, the suggested length is from 150 to 225 words.

14. KEY WORDS: Key words are technically meaningful terms or short phrases that characterize a report and may be used as index entries for cataloging the report. Key words must be selected so that no security classification is required. Identifiers, such as equipment model designation, trade name, military project code name, geographic location, may be used as key words but will be followed by an indication of technical context. The assignment of links, roles, and weights is optional.